

Statistics & Probability

ANSWER KEY



Mean, median, and mode

- Find the mean of 4, 8, 6, 10, 7.
Sum = 35. Mean = $\frac{35}{5} = 7$.
 - Find the median of 3, 9, 5, 1, 7.
Sorted: 1, 3, 5, 7, 9. Median = 5.
 - Find the mode of 2, 4, 4, 6, 8, 4, 2.
4 appears three times, more than any other value, so the mode is 4.
 - A data set has values 8, 12, 10, 15, 10, 14, 11. Find the mean, median, and mode.
Sum = 80, mean = $\frac{80}{7} \approx 11.43$. Sorted: 8, 10, 10, 11, 12, 14, 15; median (4th value) = 11. Mode = 10 (appears twice).
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Range and quartiles

- Find the range of 5, 12, 8, 20, 3.
Range = $20 - 3 = 17$.
 - For the data set 2, 4, 5, 7, 8, 9, 12, find Q_1 and Q_3 .
Median (4th value) = 7. Lower half 2, 4, 5: $Q_1 = 4$. Upper half 8, 9, 12: $Q_3 = 9$.
 - Find the interquartile range (IQR) for the same data set.
 $IQR = Q_3 - Q_1 = 9 - 4 = 5$.
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Variance and standard deviation

- Find the variance of 2, 4, 6, 8, 10, using $\sigma^2 = \frac{\sum(x - \mu)^2}{n}$.
Mean = 6. Squared deviations: 16, 4, 0, 4, 16, sum = 40. Variance = $\frac{40}{5} = 8$.
 - Find the standard deviation of the same data set, rounded to two decimal places.
 $\sigma = \sqrt{8} \approx 2.83$.
 - A data set has variance 25. State its standard deviation.
 $\sigma = \sqrt{25} = 5$.
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Frequency tables

- 11 A frequency table shows value 1 with frequency 3, value 2 with frequency 5, and value 3 with frequency 2. Find the mean.

$$\text{Sum} = 1(3) + 2(5) + 3(2) = 3 + 10 + 6 = 19. \text{ Total frequency} = 10. \text{ Mean} = \frac{19}{10} = 1.9.$$

- 12 Using the same frequency table, find the mode.

Value 2 has the highest frequency (5), so the mode is 2.

- 13 Using the same frequency table, find the median.

There are 10 values in order 1, 1, 1, 2, 2, 2, 2, 2, 3, 3. Median = $\frac{2 + 2}{2} = 2$ (average of the 5th and 6th values).

Basic probability

- 14 A fair six-sided die is rolled. Find $P(\text{number} > 4)$.

$$\text{Favorable outcomes } \{5, 6\}: P = \frac{2}{6} = \frac{1}{3}.$$

- 15 A bag contains 5 red and 3 blue marbles. Find $P(\text{blue})$.

$$P(\text{blue}) = \frac{3}{8}.$$

- 16 A card is drawn from a standard 52-card deck. Find $P(\text{king})$.

$$P(\text{king}) = \frac{4}{52} = \frac{1}{13}.$$

- 17 Two fair coins are flipped. Find $P(\text{both heads})$.

$$P = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Complementary and mutually exclusive events

- 18 If $P(A) = 0.35$, find $P(\text{not } A)$.

$$P(\text{not } A) = 1 - 0.35 = 0.65.$$

- 19 A die is rolled. Events $A = \{\text{even}\}$ and $B = \{\text{rolling a } 5\}$. Are A and B mutually exclusive?

Yes — $\{2, 4, 6\}$ and $\{5\}$ share no outcomes.

- 20 If $P(A) = 0.4$ and $P(B) = 0.25$, and A, B are mutually exclusive, find $P(A \text{ or } B)$.

$$P(A \text{ or } B) = 0.4 + 0.25 = 0.65.$$

Independent events

- 21 Two independent events have $P(A) = 0.6$ and $P(B) = 0.5$. Find $P(A \text{ and } B)$.

$$P(A \text{ and } B) = 0.6 \times 0.5 = 0.3.$$

- 22 A fair coin is flipped twice. Find $P(\text{first heads and second tails})$.
 $P = 0.5 \times 0.5 = 0.25$.
- 23 A card is drawn from a deck, replaced, then a second card is drawn. Find $P(\text{both are aces})$.
 $P = \frac{4}{52} \times \frac{4}{52} = \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$.
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Conditional probability

- 24 In a class, $P(\text{math}) = 0.7$ and $P(\text{physics and math}) = 0.4$. Find $P(\text{physics} \mid \text{math})$.
 $P(\text{physics} \mid \text{math}) = \frac{0.4}{0.7} \approx 0.571$.
- 25 A box has 10 balls: 6 red and 4 blue. Two balls are drawn without replacement. Find $P(\text{2nd is blue} \mid \text{1st is blue})$.
After removing one blue ball, 3 of the remaining 9 balls are blue: $P = \frac{3}{9} = \frac{1}{3}$.
- 26 Given $P(A) = 0.5$, $P(B) = 0.3$, $P(A \text{ and } B) = 0.15$, find $P(A \mid B)$.
 $P(A \mid B) = \frac{0.15}{0.3} = 0.5$.
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Counting: permutations and combinations

- 27 In how many ways can 4 books be arranged on a shelf?
 $4! = 24$ ways.
- 28 How many ways can a committee of 3 be chosen from 8 people (order doesn't matter)?
 $\binom{8}{3} = \frac{8!}{3!5!} = 56$.
- 29 A code uses 3 distinct digits from 0–9, where order matters. How many possible codes are there?
 $10 \times 9 \times 8 = 720$.
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The five-number summary

- 30 For the data set 6, 10, 14, 18, 22, 26, 30, find the five-number summary (minimum, Q_1 , median, Q_3 , maximum).
Median (4th value) = 18. Lower half 6, 10, 14: $Q_1 = 10$. Upper half 22, 26, 30: $Q_3 = 26$. Summary: 6, 10, 18, 26, 30.
- 31 Find the IQR for the same data set.
 $IQR = Q_3 - Q_1 = 26 - 10 = 16$.
- 32 Outlier boundaries are $Q_1 - 1.5 \times IQR$ and $Q_3 + 1.5 \times IQR$. Find both boundaries for the same data set.
Lower: $10 - 1.5(16) = 10 - 24 = -14$. Upper: $26 + 1.5(16) = 26 + 24 = 50$.

Reading data from a bar chart

- 33** The bar chart shows the number of books sold each day at a bookstore over five days.
- a** Find the total number of books sold over the five days. Total = $12 + 18 + 9 + 15 + 21 = 75$ books.
 - b** Find the mean number of books sold per day. Mean = $\frac{75}{5} = 15$ books per day.
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Word problems: statistics and probability in context

- 34** This question has two parts. A survey of 200 students found that 120 like math and 90 like science, while 50 like both subjects.
- a** Find the probability that a randomly selected student likes math or science, using
 $P(M \cup S) = P(M) + P(S) - P(M \cap S)$. $P(M) = \frac{120}{200} = 0.6$, $P(S) = \frac{90}{200} = 0.45$, $P(M \cap S) = \frac{50}{200} = 0.25$.
 $P(M \cup S) = 0.6 + 0.45 - 0.25 = 0.8$.
 - b** Find the probability that a student likes neither subject. $P(\text{neither}) = 1 - 0.8 = 0.2$.
- 35** This question has two parts. A quality-control inspector finds that 2% of items from a production line are defective. A random sample of 3 items is tested, and defects occur independently.
- a** Find the probability that all three items are defective. $P(\text{all defective}) = 0.02^3 = 0.000008$.
 - b** Find the probability that none of the three items are defective.
 $P(\text{none defective}) = 0.98^3 \approx 0.9412$.